DQLab

BBCA Stock Performance Analysis (2014–2024)

Data Analyst Portofolio Project

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Introduction

Background

The stock market offers attractive return potential but is exposed to volatility and systemic risk.

BBCA (Bank Central Asia) is:

- One of Indonesia's largest blue-chip banks
- Known for strong fundamentals and consistent profitability
- Considered a defensive growth stock
- Widely held by long-term investors

This analysis evaluates BBCA's performance, risk profile, and return behavior over the past 10 years.

Introduction

Analysis Objectives

This analysis aims to:

- Measure BBCA's long-term growth performance using CAGR
- Evaluate risk exposure through volatility and drawdown analysis
- Assess yearly return consistency
- Analyze daily return distribution characteristics
- Measure risk-adjusted performance using Sharpe Ratio
- Examine rolling 1-year momentum using Rolling 252

Data Overview

Data Description

Data Source:
Yahoo Finance – BBCA.JK Historical Data
https://finance.yahoo.com/quote/BBCA.JK/history
Time Period:
December 2014 – December 2024
Frequency:
Daily trading data
Variables:
Date, Open, High, Low, Close, Volume
Total Observations:
10 years of daily market data

DOLab

live Class

Data Overview

Data Processing

Tools

Google Colab

Python Programming Language

Google Looker Studio

Data Overview

Data Processing

Data Cleaning

- Converted Date column to datetime format
- Sorted data chronologically
- Checked and handled missing values
- Calculated daily returns

Feature Engineering

Created additional analytical metrics:

- Moving Average (50-day & 200-day)
- 30-Day Rolling Volatility
- Drawdown
- Compound Annual Growth Rate (CAGR)
- Annual Return
- Rolling 252 (1-Year Momentum)
- Sharpe Ratio

Data Cleaning

Before and After

Raw Data

index	Date	Close	High	Low	Open	Volume
0	2014-12-08 00:00:00+00:00	2655	2680	2645	2655	37435500
1	2014-12-09 00:00:00+00:00	2650	2680	2630	2660	38254000
2	2014-12-10 00:00:00+00:00	2660	2675	2630	2630	32998000
3	2014-12-11 00:00:00+00:00	2655	2675	2645	2660	54459500
4	2014-12-12 00:00:00+00:00	2650	2665	2640	2640	50302000

Cleaned and added

index	Date	Close	High	Low	Open	Volume	daily_return	MA_50	MA_200	volatility_30	cum_max	drawdown	Year	rolling_252	CAGR	Sharpe_Ratio	Max_Gain	Max_Loss	Annual_Return
2471	2024-12-02	9750.0	10050.0	9750.0	10000.0	87879600	-0.025000000000000022	10371.0	10017.125	0.013635308681747784	10950.0	-0.1095890410958904	2024	0.12204429099764302	0.14415697253368087	0.7131007230621601	0.17333333333333334	-0.07913669064748197	0.0851063829787233
2472	2024-12-03	10200.0	10225.0	9900.0	10025.0	126087100	0.04615384615384621	10356.0	10020.375	0.016310426832386492	10950.0	-0.0684931506849315	2024	0.16253241477188574	0.14415697253368087	0.7131007230621601	0.17333333333333334	-0.07913669064748197	0.0851063829787233
2469	2024-11-28	10000.0	10075.0	9925.0	9975.0	57443500	0.0	10409.5	10012.625	0.013868733183317916	10950.0	-0.0867579908675799	2024	0.12759984655319856	0.14415697253368087	0.7131007230621601	0.17333333333333334	-0.07913669064748197	0.0851063829787233
2470	2024-11-29	10000.0	10100.0	9950.0	10000.0	76886000	0.0	10391.5	10015.125	0.013025132490224233	10950.0	-0.0867579908675799	2024	0.12759984655319856	0.14415697253368087	0.7131007230621601	0.17333333333333334	-0.07913669064748197	0.0851063829787233
2468	2024-11-26	10000.0	10250.0	10000.0	10250.0	62310100	-0.017199017199017175	10422.0	10010.25	0.014055985074284098	10950.0	-0.0867579908675799	2024	0.13038536187353278	0.14415697253368087	0.7131007230621601	0.17333333333333334	-0.07913669064748197	0.0851063829787233

Python Link:
https://colab.research.google.com/drive/1olbrdLQ13Cp0fD-_bSJu0vjT0jtheRE0?usp=sharing

Executive Summary

Key Performance Metric

CAGR	Volatility
14.42%	1.34%
Sharpe Ratio	
0.71	

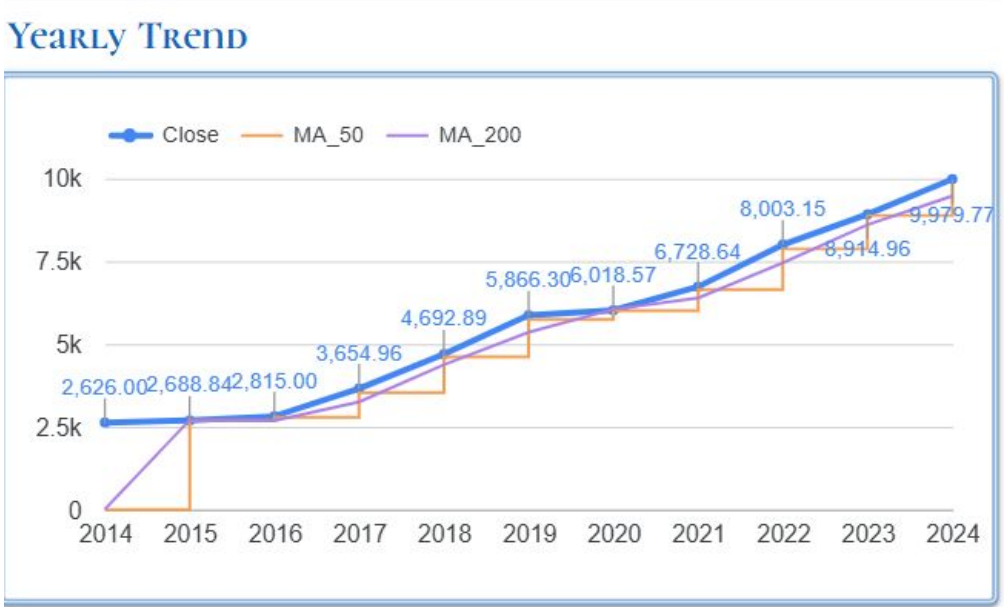
Highest Drop	Max Price
-35.56%	10950
Average Daily...	**Annual Return**
0.06%	15.39%

Insights

- BBCA demonstrates strong long-term growth with double-digit CAGR.
- Risk exposure remains moderate for a large-cap stock.
- Significant drawdown occurred during crisis periods but recovery was relatively fast.
- Risk-adjusted return is reasonable based on Sharpe Ratio.

Overview

Long-Term Trend Analysis

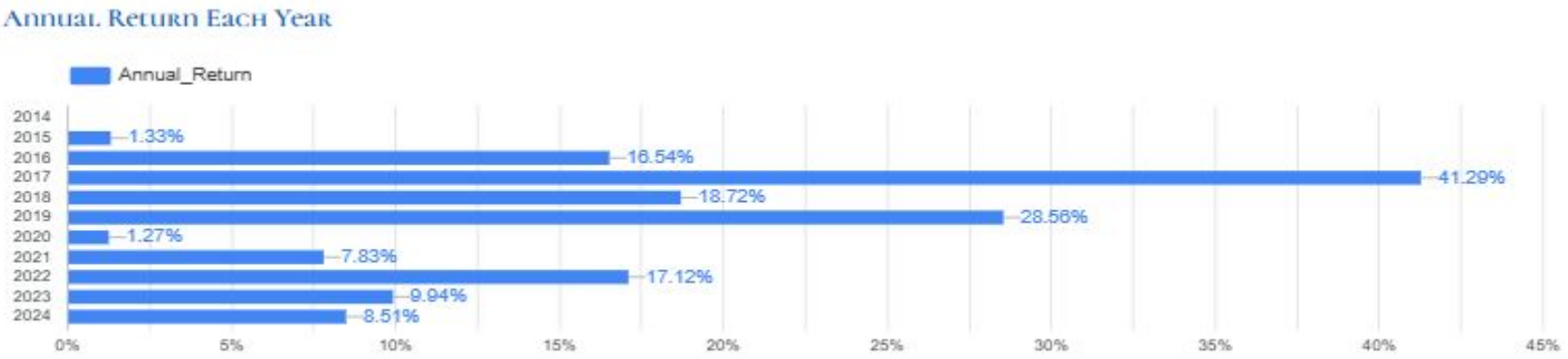


Key Observations

- Clear long-term upward trend from 2014 to 2024
- Significant correction during 2020 market crisis
- Strong recovery following drawdown
- Moving averages confirm sustained bullish structure

Trend Analysis

Annual Performance Analysis

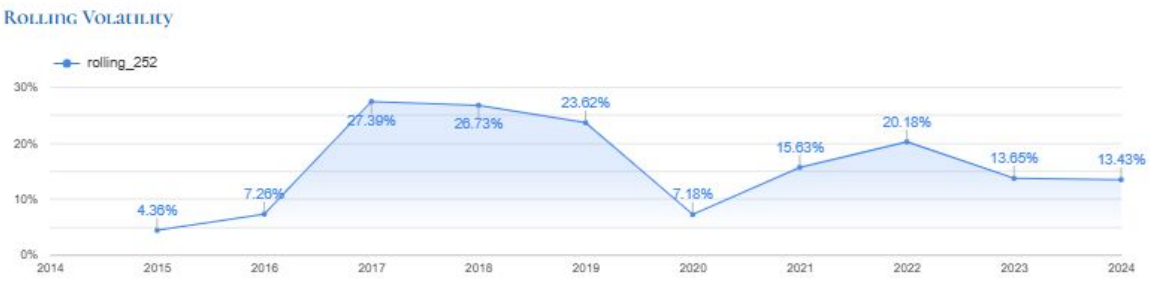


Key Observations

- Strong double-digit returns in expansion years
- Peak annual return observed in 2017 (~41%)
- Weakest performance during early years / crisis periods
- Majority of years delivered positive returns

Trend Analysis

Rolling 252 1-Year Momentum Analysis



Key Observations

- Momentum peaked at approximately ~27% during strong expansion phases
- Momentum declined significantly during crisis periods
- Recovery phases show rapid improvement in trailing returns
- Rolling performance highlights market cycle dynamics

Risk Analysis

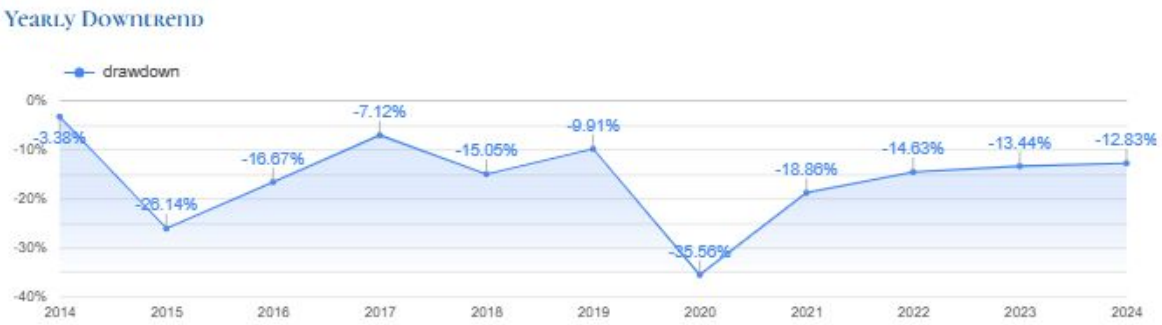
Yearly Downtrend

Max_Gain

17.33%

Max_Loss

-7.91%

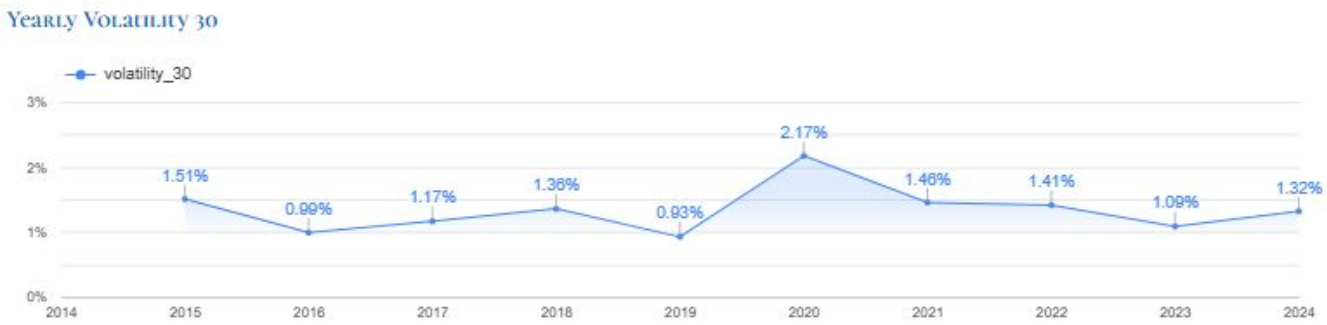


Key Observations

- Maximum drawdown reached approximately **-35%**
- Largest decline occurred during the 2020 crisis
- Recovery period followed within 1–2 years
- Drawdowns were episodic rather than persistent

Risk Analysis

Volatility Analysis

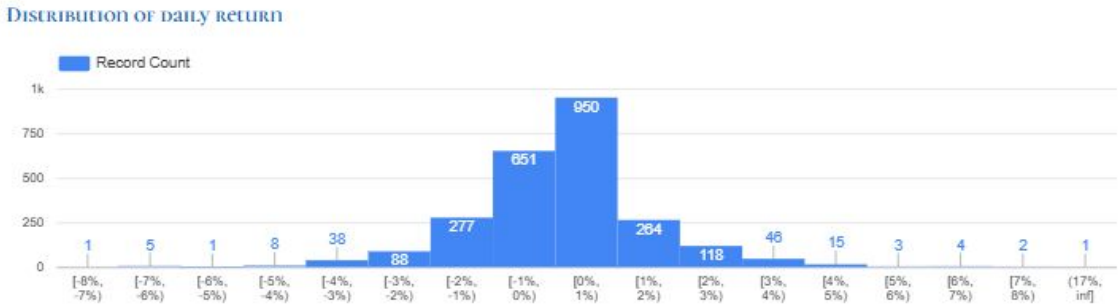


Key Observations

- Volatility spiked during the 2020 crisis period
- Average volatility remains around ~1–1.5%
- Volatility normalizes after market shocks
- Large-cap characteristics reflected in relatively stable movement

Performance and Distribution

Return Distribution Analysis



Key Observations

- Majority of daily returns fall between -2% and +2%
- Distribution is relatively centered around zero
- Extreme positive and negative returns are rare
- Limited downside tail risk observed

Summary

Key Findings

- BBCA delivered strong long-term growth with a 10-year CAGR of ~14%.
- Annual returns were predominantly positive, indicating performance consistency.
- Maximum drawdown of ~35% occurred during crisis periods but recovery was relatively fast.
- Volatility remained moderate for a large-cap stock, reflecting controlled risk exposure.
- Sharpe Ratio of 0.71 suggests reasonable risk-adjusted returns.

Summary

Investment Implications & Recommendations

- BBCA is suitable for long-term investors seeking stable growth.
- Strategic accumulation during major drawdowns may optimize long-term returns.
- Monitoring rolling 1-year momentum can help identify entry and exit timing.
- Despite moderate volatility, diversification remains important to manage systemic risk.
- Investors should remain cautious during macroeconomic shocks that may trigger temporary drawdowns.

Summary

Limitations & Future Improvements

Limitations

- Dividend returns were not included in total return calculation.
- No benchmark comparison (e.g., IHSG index) was performed.
- Macroeconomic factors such as interest rates and inflation were not incorporated.
- Analysis is based solely on historical price behavior.

Future Improvements

- Compare BBCA performance against IHSG or banking sector index.
- Include dividend-adjusted total return analysis.
- Incorporate macroeconomic indicators for deeper insight.
- Extend analysis with predictive modeling or forecasting.

Conclusion

- BBCA demonstrates sustainable long-term growth with a 10-year CAGR of ~14%.
- Risk exposure remains moderate despite temporary crisis-driven drawdowns.
- Performance consistency and rolling momentum indicate structural strength.
- Overall, BBCA presents an attractive balance between growth and risk for long-term investors.

Thanks!

BBCA Stock Performance Analysis (2014–2024)

8 Dec 2014 - 3 Dec 2024

Highest Drop

-12.83%

Max Price

10950

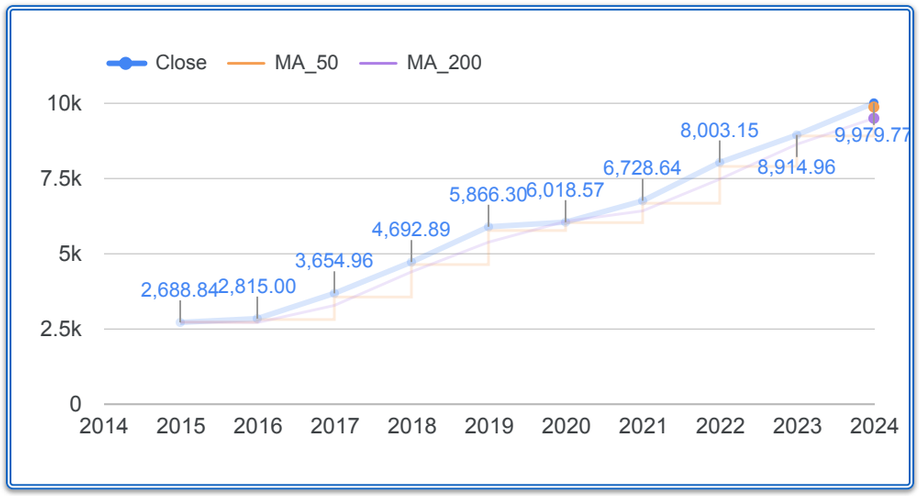
Average Daily...

0.05%

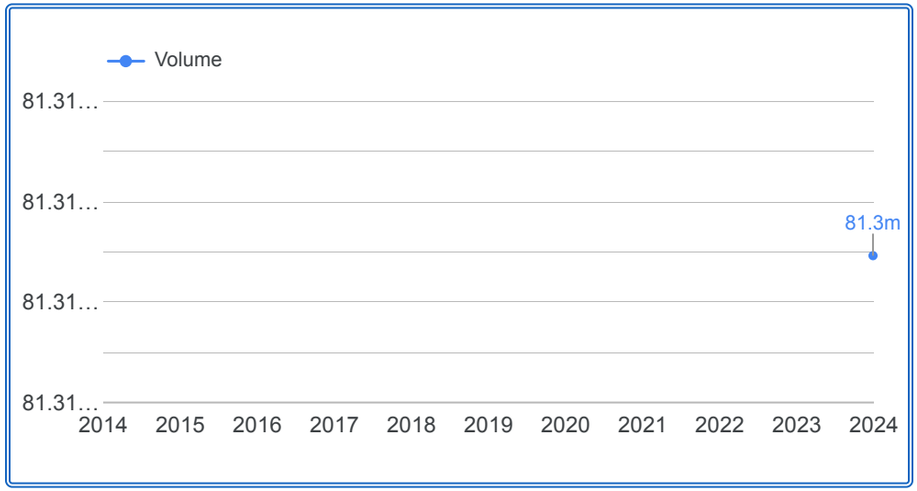
Annual Return

8.51%

Yearly Trend



Yearly Volume



CAGR

14.42%

Sharpe Ratio

0.71

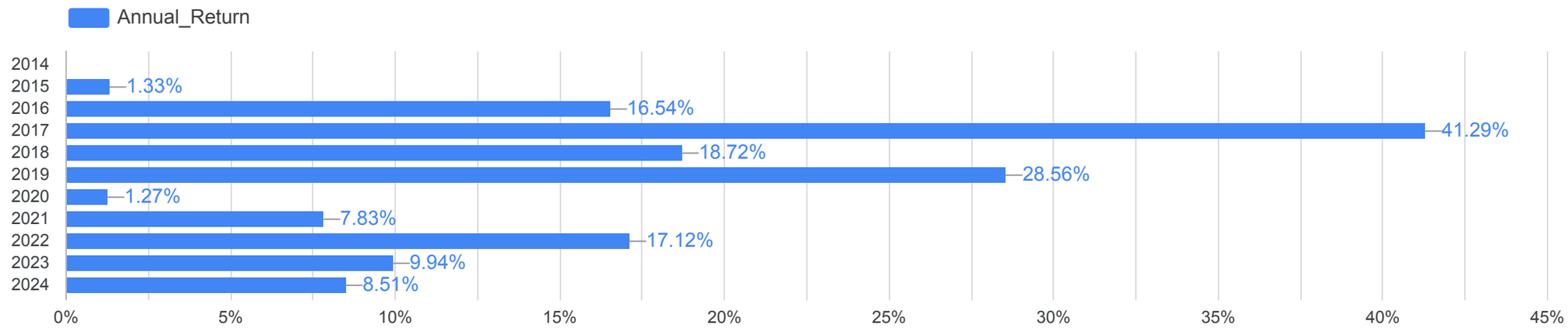
Volatility

1.32%

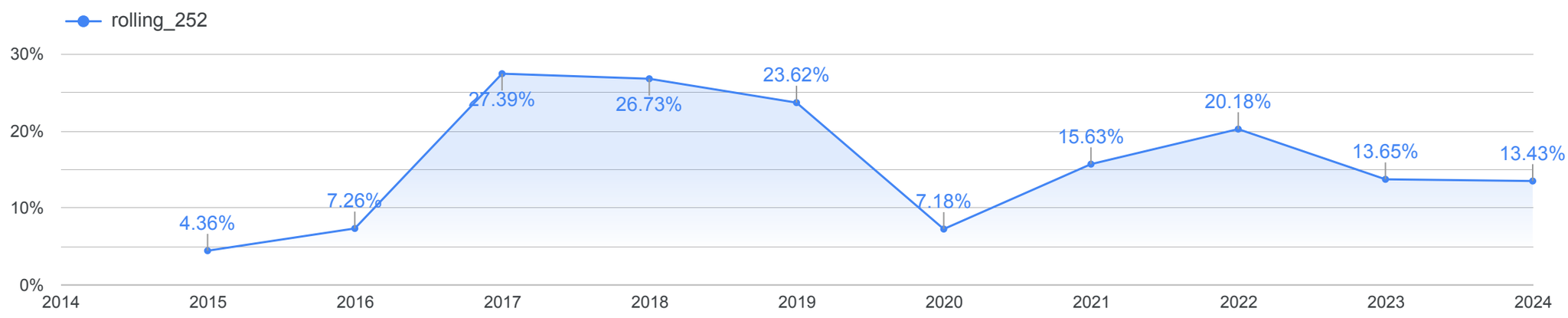
BBCA Stock Performance Analysis (2014–2024)

Trend Analysis

Annual Return Each Year



Rolling Volatility



BBCA Stock Performance Analysis (2014–2024)

Risk Analysis

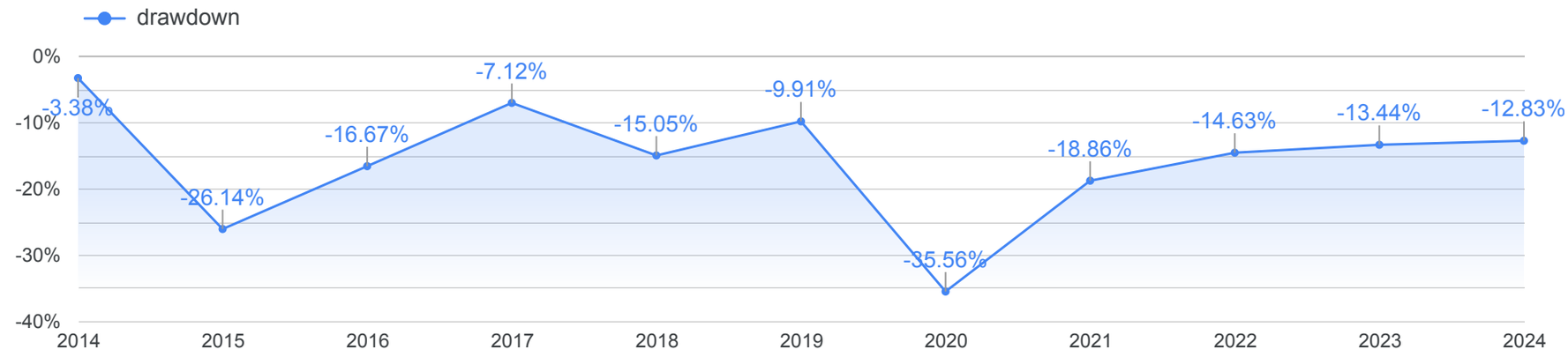
Max_Gain

17.33%

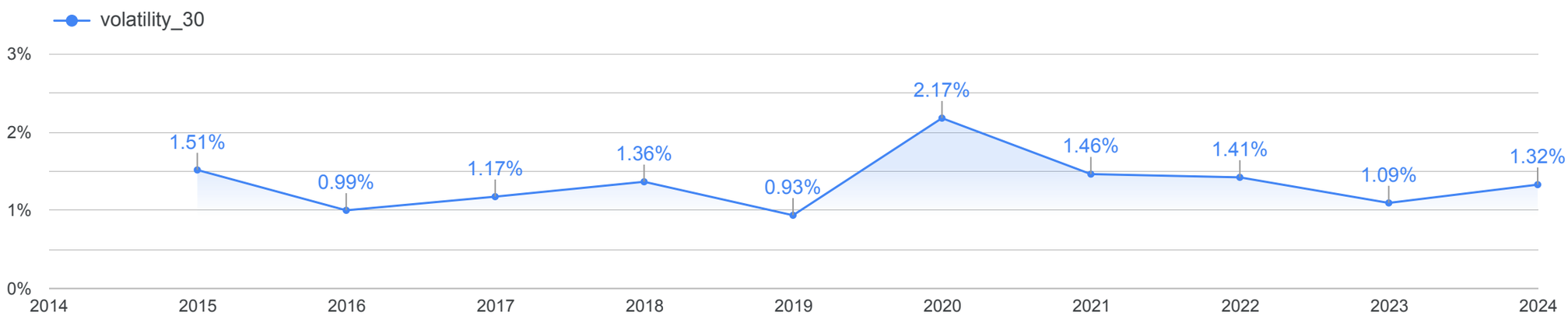
Max_Loss

-7.91%

Yearly Downtrend



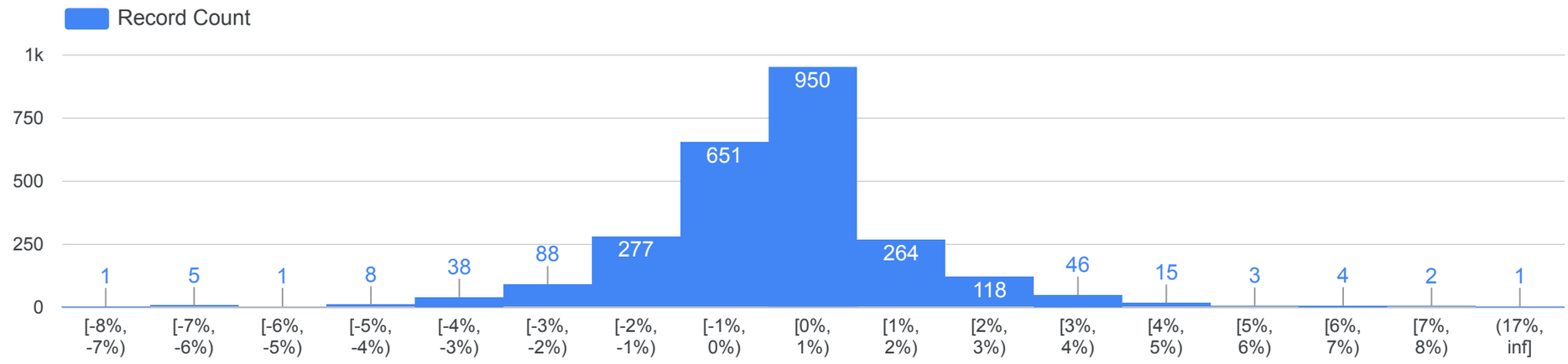
Yearly Volatility 30



BBCA Stock Performance Analysis (2014–2024)

Performance and Distribution

Distribution of Daily Return



Yearly Summary

	Year ▲	Annual_Return	volatility_30
1.	2014	null	null
2.	2015	1.33%	1.51%
3.	2016	16.54%	0.99%
4.	2017	41.29%	1.17%
5.	2018	18.72%	1.36%
6.	2019	28.56%	0.93%
7.	2020	1.27%	2.17%
8.	2021	7.83%	1.46%
9.	2022	17.12%	1.41%
10.	2023	9.94%	1.09%
11.	2024	8.51%	1.32%